Exact simulation of Lévy random-bridges without the martingale $\{\mathbf{M}_t\}$

Gamma and Poisson cases, with a proposed head-to-head experiment

Working note for internal review

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Abstract

Section 4 of the manuscript derives the non-anticipative representation $\boldsymbol{\xi}_t^{\boldsymbol{x}} \overset{\text{law}}{=} \int_0^t \frac{\mathbb{E}[\boldsymbol{Y}|\boldsymbol{\xi}_s^{\boldsymbol{x}}]-\boldsymbol{\xi}_s^{\boldsymbol{x}}}{T-s}\,\mathrm{d}s + \mathbf{M}_t$ and states that a practical algorithm awaits an explicit form for the martingale $\{\mathbf{M}_t\}$. This note observes that the simulation does not require $\{\mathbf{M}_t\}$ at all. Sampling the *bridge transition kernel* $\nu_{s,t,T}$ — already derived in the manuscript — advances the state exactly, with no time-discretisation error and no terminal singularity. For the gamma process the kernel is a scaled Beta law; for the Poisson process it is a Binomial thinning. Both are elementary to simulate. This converts Section 4 from a theoretical outlook into a working algorithm, and unifies the Gaussian, gamma and Poisson cases under one sampler. **Please check Propositions 2 and 3; the algebra is elementary but I would like it verified independently before we build on it.**

1 Setting and the point of departure

Let $\{\mathbf{Z}_t\}_{t\in\mathcal{T}}$, $\mathcal{T} = [0,T]$, be a Lévy process with marginal law f_u, and let $\{\boldsymbol{\xi}_t^{\boldsymbol{x}}\}$ be the $\boldsymbol{\Phi}$-initialised Lévy random-bridge to $\boldsymbol{\Psi}$ with $\boldsymbol{\xi}_0^{\boldsymbol{x}} = \boldsymbol{x}$. Assume coordinate-wise independence throughout, so that all statements below are one-dimensional and apply coordinate-by-coordinate.

The manuscript establishes (Cor. 4.4 and Sec. 4.1) that for $0 \le s < t < T$ the conditional point-to-point bridge $\boldsymbol{\xi}^{\boldsymbol{x},\boldsymbol{y}}$ has transition density

$$\mathbb{P}\big(\boldsymbol{\xi}_t^{\boldsymbol{x},\boldsymbol{y}} \in \mathrm{d}z \,\big|\, \boldsymbol{\xi}_s^{\boldsymbol{x},\boldsymbol{y}} = r\big) \;=\; \frac{f_{t-s}(z-r)\, f_{T-t}(y-z)}{f_{T-s}(y-r)}\,\mathrm{d}z \;\triangleq\; \nu_{s,t,T}(r,z,y)\,\mathrm{d}z. \tag{1}$$

Remark 1 (The observation)**.** *The representation involving* $\{\mathbf{M}_t\}$ *is the canonical decomposition of the bridge into a drift plus a compensated remainder. It is the natural object if one intends to* integrate an SDE *by a time-stepping scheme, because* $\mathbf{M}_t$ *is precisely what such a scheme must inject at each step. It is not needed if one instead* samples (1) directly*. Since* (1) *is the exact law of* $\boldsymbol{\xi}_t$ *given* $\boldsymbol{\xi}_s$ *and the endpoint, drawing from it advances the process with no discretisation error whatsoever, for any step size. The martingale is an artefact of the integrator, not of the process.*

In the generative setting the endpoint $\boldsymbol{y}$ is unknown and is replaced by the network's estimate $\hat{\boldsymbol{y}} = f^*(\boldsymbol{\xi}_s, s; \Theta^*) \approx \mathbb{E}[\boldsymbol{Y} \mid \boldsymbol{\xi}_s]$. This is the same substitution that DDPM/DDIM ancestral sampling makes, and it is discussed in Section 5.

Throughout we use normalised time $u = t/T \in [0,1]$ and write the horizon as 1.

2 Gamma random-bridges

Let $\{Z_t\}$ be a gamma subordinator with $\mathbb{E}[Z_t] = t$ and shape rate $\kappa > 0$, so that increments satisfy $Z_t - Z_s \sim \mathrm{Gamma}\big(\kappa(t-s), \kappa\big)$ with density

$$f_u(w) = \frac{\kappa^{\kappa u}}{\Gamma(\kappa u)}\, w^{\kappa u - 1} e^{-\kappa w}, \qquad w > 0. \tag{2}$$

Proposition 2 (Gamma bridge transition)**.** *For $0 \le s < t < 1$ and $r < y$,*

$$\xi_t \;=\; \xi_s + (y - \xi_s)\,\beta, \qquad \beta \sim \mathrm{Beta}\big(\kappa(t-s),\, \kappa(1-t)\big), \tag{3}$$

independently of ξ_s. The rate parameter κ enters only through the Beta shape parameters; the scale of the gamma process cancels identically.

Proof. Put $a = \kappa(t-s)$, $b = \kappa(1-t)$, so $a + b = \kappa(1-s)$. Substituting the gamma densities into (1),

$$\begin{aligned}\nu_{s,t,1}(r,z,y) &= \frac{\dfrac{\kappa^a}{\Gamma(a)}(z-r)^{a-1}e^{-\kappa(z-r)} \cdot \dfrac{\kappa^b}{\Gamma(b)}(y-z)^{b-1}e^{-\kappa(y-z)}}{\dfrac{\kappa^{a+b}}{\Gamma(a+b)}(y-r)^{a+b-1}e^{-\kappa(y-r)}} \\ &= \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \cdot \frac{(z-r)^{a-1}(y-z)^{b-1}}{(y-r)^{a+b-1}},\end{aligned} \tag{4}$$

the exponential factors cancelling exactly because $(z-r)+(y-z) = (y-r)$, and the powers of κ cancelling because $a+b$ is the total shape. Substituting $z = r + (y-r)\beta$ with $\mathrm{d}z = (y-r)\,\mathrm{d}\beta$ and $\beta \in (0,1)$,

$$\nu\ \mathrm{d}z = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)}\, \beta^{a-1}(1-\beta)^{b-1}\ \mathrm{d}\beta, \tag{5}$$

which is the $\mathrm{Beta}(a,b)$ density. □

Corollary 3 (Training marginal)**.** *With $\xi_0 = x$ and terminal y, the marginal used for training is $\xi_s = x + (y-x)\,\beta$, $\beta \sim \mathrm{Beta}(\kappa s, \kappa(1-s))$.*

Corollary 4 (Moments)**.** $\mathbb{E}[\xi_t \mid \xi_s, y] = \xi_s + (y - \xi_s)\frac{t-s}{1-s}$ *and*

$$\mathrm{Var}[\xi_t \mid \xi_s, y] = (y-\xi_s)^2\, \frac{(t-s)(1-t)}{(1-s)^2\big(\kappa(1-s)+1\big)}. \tag{6}$$

The conditional mean is *identical* to the Gaussian case and reproduces the linear interpolation property of manuscript eq. (4.x); only the fluctuation law differs. As $\kappa \to \infty$ the variance vanishes and the bridge degenerates to the deterministic linear interpolant.

Remark 5 (Support)**.** *A gamma subordinator is non-decreasing, so the construction requires $\xi_s < y$ coordinate-wise. The gamma bridge is therefore a transport between* ordered *laws, and its paths remain in $[\xi_s, y]$ almost surely. This is the order-preserving regime motivated in the Introduction of the manuscript.*

3 Poisson random-bridges

Let $\{Z_t\}$ be a Poisson process of rate λ, so $Z_t - Z_s \sim \mathrm{Poisson}(\lambda(t-s))$ with mass function $f_u(k) = e^{-\lambda u}(\lambda u)^k/k!$, $k \in \mathbb{N}_0$.

Proposition 6 (Poisson bridge transition)**.** *For $0 \le s < t < 1$ and integers $r \le y$, writing $m = y - r$,*

$$\xi_t \;=\; \xi_s + \mathrm{Binomial}\!\left(y - \xi_s,\; \frac{t-s}{1-s}\right). \tag{7}$$

The rate λ cancels identically.

Proof. With $j = z - r$ and $m = y - r$, the discrete analogue of (1) gives

$$\begin{aligned}\mathbb{P}(\xi_t = z \mid \xi_s = r, Y = y) &= \frac{e^{-\lambda(t-s)}\frac{(\lambda(t-s))^j}{j!}\; e^{-\lambda(1-t)}\frac{(\lambda(1-t))^{m-j}}{(m-j)!}}{e^{-\lambda(1-s)}\frac{(\lambda(1-s))^m}{m!}} \\ &= \binom{m}{j}\frac{(t-s)^j(1-t)^{m-j}}{(1-s)^m} \;=\; \binom{m}{j}p^j(1-p)^{m-j}, \end{aligned} \tag{8}$$

with $p = \frac{t-s}{1-s}$, since the exponentials combine as $e^{-\lambda(t-s)}e^{-\lambda(1-t)} = e^{-\lambda(1-s)}$ and the powers of λ cancel as $\lambda^j\lambda^{m-j} = \lambda^m$. □

Corollary 7. *Training marginal:* $\xi_s = x + \mathrm{Binomial}(y - x,\, s)$.
Moments: $\mathbb{E}[\xi_t \mid \xi_s, y] = \xi_s + (y - \xi_s)\frac{t-s}{1-s}$ *and* $\mathrm{Var}[\xi_t \mid \xi_s, y] = (y - \xi_s)\frac{(t-s)(1-t)}{(1-s)^2}$.

Again the conditional mean is the common linear interpolant. The Poisson bridge is the classical binomial thinning; the process remains on the integer lattice at every step, by construction and not by rounding.

4 The unified sampler

Writing $\hat{y} = f^*(\xi_s, s; \Theta^*)$ for the network's endpoint estimate, all three cases share the update

$$\xi_u \;=\; \underbrace{\xi_s + (\hat{y} - \xi_s)\tfrac{u-s}{1-s}}_{\text{common drift}} \;+\; \underbrace{(\text{driver-specific fluctuation})}, \tag{9}$$

with the fluctuation given in Table 1. In every case the fluctuation degenerates as $u \to 1$ and the scheme lands on $\hat{y}$ exactly, so no boundary offset ϵ and no $1/(T-t)$ singularity arise.

Driver	Exact transition $\xi_s \to \xi_u$	Support	Parameter
Brownian σW	$\mathcal{N}\big(\text{drift},\; \gamma^2\frac{(u-s)(1-u)}{1-s}\big)$	$\mathbb{R}$	$\gamma^2 = \sigma^2 T$
Gamma	$\xi_s + (\hat{y} - \xi_s)\,\mathrm{Beta}\big(\kappa(u-s), \kappa(1-u)\big)$	$[\xi_s, \hat{y}]$	κ
Poisson	$\xi_s + \mathrm{Bin}\big(\hat{y} - \xi_s,\; \frac{u-s}{1-s}\big)$	$\{\xi_s, \ldots, \hat{y}\} \subset \mathbb{N}_0$	—

Table 1: One algorithm, three drivers. The drift is common; only the fluctuation law changes. Training samples the corresponding marginal from (x, y) in one shot.

Remark 8 (Gaussian case, for completeness)**.** *For $\mathbf{Z}_t = \sigma\mathbf{W}_t$ the same substitution into (1) gives the Brownian bridge on $[s, 1]$ pinned at (s, ξ_s) and $(1, \hat{y})$, i.e. the first row of Table 1. This has been implemented and verified numerically: on a one-dimensional problem with $\mathbb{E}[Y \mid \xi_s]$ available in closed form, it attains the Monte-Carlo floor in 1-Wasserstein distance, whereas the deterministic Euler map plateaus at $6.9\times$ that floor.*

5 Caveat: the point-mass substitution

The exactness in Propositions 2 and 6 is exactness *given the endpoint*. In generation we replace the posterior $\pi_s(\mathrm{d}y) = \mathbb{P}(\boldsymbol{Y} \in \mathrm{d}y \mid \boldsymbol{\xi}_s)$ by a point mass at its mean $\hat{y}$. The sampler is therefore exact in the SDE-discretisation sense and approximate in the posterior sense, under-dispersing by $\mathrm{Var}[\boldsymbol{Y} \mid \boldsymbol{\xi}_s]$ — exactly the quantity Prop. 2.13 of the manuscript shows to be a Loewner-order supermartingale vanishing at T. Two remarks:

1. This is the same approximation DDPM and DDIM make, so it is not a new weakness relative to the comparison set.

2. It is removable in principle: a second network head estimating $\mathrm{Var}[\boldsymbol{Y} \mid \boldsymbol{\xi}_s]$ would allow the dropped dispersion to be reinstated. Prop. 2.13 supplies the theoretical object; whether this helps empirically is open.

A further practical point for the Poisson case: $\hat{y}$ is real-valued while $\mathrm{Bin}(\cdot)$ requires an integer count, so $\hat{y} - \xi_s$ must be rounded or the head must emit an integer-valued estimate. This is the one place rounding re-enters and it deserves care.

6 Proposed head-to-head experiment

The framework's generality is currently asserted rather than demonstrated, which is the substance of reviewer point 7. The generality cannot win on MNIST or CIFAR-10: those are unconstrained real-valued supports, where the Brownian driver is the correct choice and the framework can at best tie a well-tuned flow-matching or diffusion model. It can only win where the Gaussian case is *structurally* wrong.

Claim to target. *On count-valued data, a Poisson random-bridge generates on the integer lattice exactly, at every sampling step, whereas Gaussian diffusion and flow matching must model a continuous relaxation and round — and the rounding costs measurably more at low NFE.*

Design. A single controlled comparison, all arms sharing architecture, optimiser, EMA, batch size, training steps and evaluation code; only the transport changes.

1. **Poisson bridge** (ours, Prop. 6).

2. **Gaussian bridge** (ours, wrong driver) — the internal ablation that isolates the driving process from everything else in the framework.

3. **Flow matching** with linear path, then rounding — the modern external baseline.

4. **Gaussian diffusion** (DDPM) on the relaxation, then rounding.

Data. Two tiers. *(i) Fast and self-contained:* photon-limited images. Take MNIST/CIFAR-10 intensities, fix a mean photon budget (e.g. 10 counts per pixel) and draw Poisson counts; the task is to generate realistic count images. This is cheap, reproducible and requires no new data pipeline. *(ii) Real application:* single-cell RNA-seq count matrices (e.g. 10x PBMC). Counts here are integer, sparse and over-dispersed, and Poisson/negative-binomial is the *domain-canonical* noise model — so the Poisson driver is the principled choice rather than an opportunistic one. Generative modelling of scRNA-seq is an active area with established metrics.

Metrics.

1. **Support violation rate**: fraction of generated entries outside $\mathbb{N}_0$ before repair. Ours is identically zero *by construction*; every baseline is positive. This is structural, not a tuning artefact.

2. **Distributional fidelity after the baselines' repair step**: FID for images; for scRNA-seq the standard battery (marginal count histograms, mean–variance relation, sparsity/zero-inflation rate, nearest-neighbour overlap).

3. **Quality versus NFE**, since the claimed advantage is in the low-step regime, reported against wall-clock as well as step count.

Honest risks. Two, which we should confront rather than discover in review. First, constraint-respecting continuous methods exist — Dirichlet and simplex flow matching, and discrete diffusion (D3PM, SEDD) — and a credible claim must include at least one of them, not only vanilla flow matching. Second, if the rounding penalty turns out to be small, the support-violation argument is technically correct but uninteresting; we should measure that penalty early and drop the line of attack if it is negligible.

7 Numerical verification

Both propositions were checked against the defining construction — simulate the driving process's independent increments, condition on the total, and compare the conditional law of the first increment with the claimed law. The identities under test are not used to generate the data. Script: `notes/verify_derivations.py`, 4×10^6 draws per cell, float64.

	(κ or λ, s, t)	discrepancy	MC floor	mean	$\frac{t-s}{1-s}$	var vs theory
Gamma	(3, 0.5, 0.9)	KS 0.00063	0.00091	0.79985	0.80000	0.064017/0.064000
Gamma	(20, 0.2, 0.5)	KS 0.00056	0.00098	0.37499	0.37500	0.013788/0.013787
Gamma	(20, 0.9, 0.95)	KS 0.00120	0.00109	0.49969	0.50000	0.083338/0.083333
Poisson	(7, 0.5, 0.9)	TVD 0.00054	—	3.1995	3.2000	0.6399/0.6400
Poisson	(30, 0.2, 0.5)	TVD 0.00255	—	8.9997	9.0000	5.6107/5.6250

Table 2: Representative cells; all eight gamma and eight Poisson cells tested agree with theory at the Monte-Carlo floor. The column $\frac{t-s}{1-s}$ confirms the common drift of (9) independently in both families.

Remark 9 (Implementation caveat, near-terminal gamma)**.** *When $\kappa(1-t) \ll 1$ the Beta law concentrates so sharply at 1 that $(\xi_t - \xi_s)/(y - \xi_s)$ rounds to exactly 1 in floating point — at $\kappa = 3$, $t = 0.999$ this happens in 89.9% of draws. The behaviour is correct (the sampler lands on the endpoint) but an implementation should special-case the final step rather than rely on the rounding, as the Gaussian implementation already does.*

8 Summary of what needs checking

1. Proposition 2: the Beta identity, in particular the claim that the exponential and κ factors cancel identically. (Verified numerically; the algebra still wants a second pair of eyes.)

2. Proposition 6: the Binomial identity and the cancellation of λ. (Likewise verified numerically.)

3. Remark 1: the assertion that sampling $\nu_{s,t,T}$ directly is legitimate and makes $\{\mathbf{M}_t\}$ unnecessary for simulation. This is the load-bearing claim; everything else is elementary algebra.

4. Whether Corollary 4 and its Poisson analogue are consistent with the conditional-expectation interpolation property already proved in the manuscript (they appear to be, which is a useful cross-check).